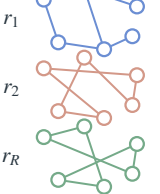


Input graph

$$\mathcal{G} = (\mathcal{V}, \{\mathcal{E}_r\}, \mathbf{X})$$



Node attributes



Label-free views

$$\mathbf{X}^p, \mathbf{X}^h$$

Supervision

0 1
 y_L only

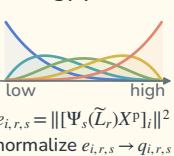
No labels needed at inference

Dataflow

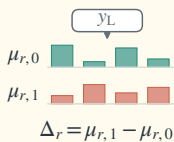
→ Forward
--- Fixed
----- Train only

(a) Orientation estimation

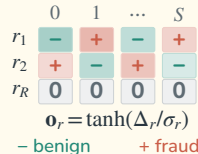
Energy profiles



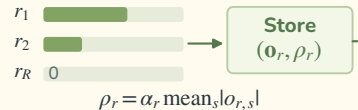
Class contrast



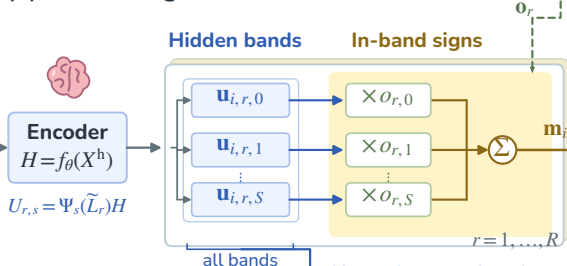
Orientation



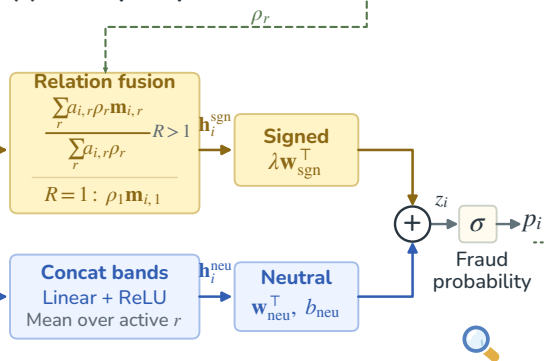
Frozen reference
Reliability & storage



(b) In-band signed residual



(c) Dual-path prediction



(d) Training objectives

